# Intelligent Valuation AI

Ajman Real Estate Intelligence — Complete Project Guide

| | | | |
|---|---|---|---|
| **Version** | **July** | **Land Management** | **e-rent-** |
| **1.0** | **2026** | **Department** | **ai.vercel.app** |
| Document | Prepared | Ajman, UAE | Live platform |

# Contents

Everything about the platform, A to Z — what it is, how it works, how to run it, and how to keep it healthy.


| | |
|---|---|
| 1 | Executive summary |
| 2 | System architecture — how the pieces fit |
| 3 | Project structure — frontend and backend |
| 4 | Valuation methodology |
| 5 | The dashboard, page by page |
| 6 | Data operations — refreshing the data |
| 7 | Automation — schedule and cloud refresh |
| 8 | Accounts, secrets and handover checklist |
| 9 | Troubleshooting |
| 10 | Maintenance, costs and honest limits |
| 11 | Glossary |

# 1. Executive summary

What the platform is and what it delivers.

**Intelligent Valuation AI** is an internal real estate intelligence platform for Ajman. It automatically collects live property listings from the three major UAE portals — **Bayut, Dubizzle and Property Finder** — and turns them into clear, defensible valuations for **land plots** and **commercial units** across **54 Ajman areas**.

## What the user gets

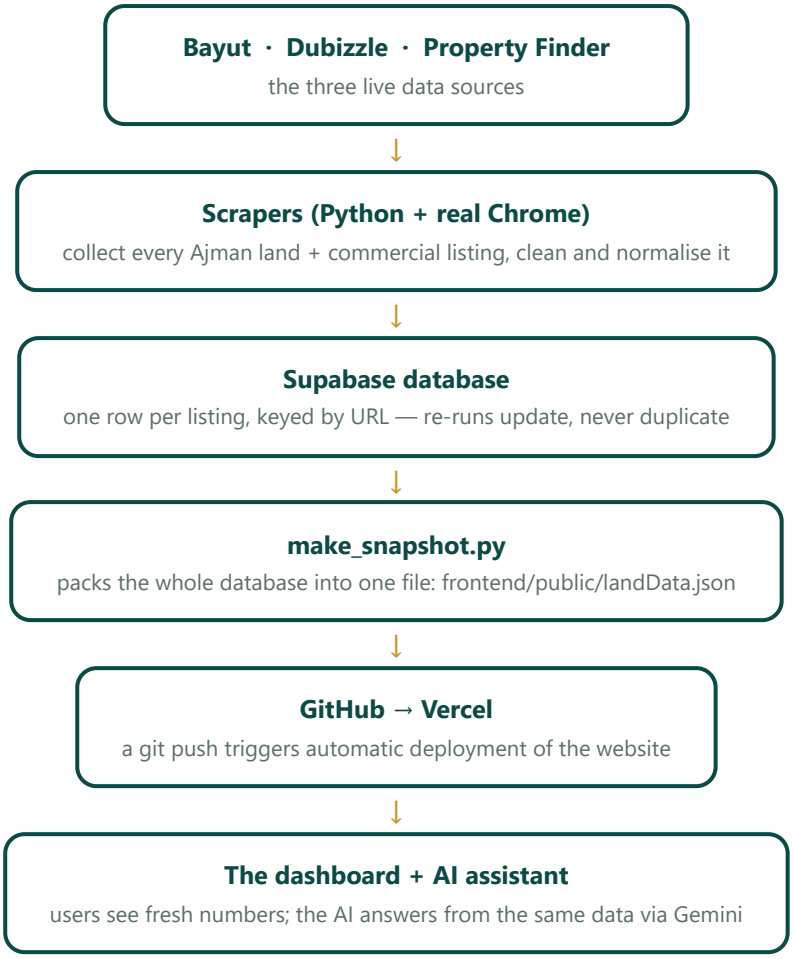
- Annual rent ranges and averages per area, for land and for every commercial unit type (offices, shops, warehouses, showrooms, buildings and more)
- Interactive map (standard and satellite), sortable listings with links to the original ads, trends, insights and area comparisons
- An AI assistant (text and voice, English and Arabic) that answers questions grounded in the live data
- Rent estimator, ROI calculator and printable reports
- Fully bilingual interface (English / العربية) with full right-to-left support
- Installable on any phone as an app icon — no app store needed (PWA)

## Key numbers (July 2026)

| Metric | Value |
|---|---|
| Listings in database | 9,250+ (deduplicated by listing URL) |
| Land listings | ~3,670 (sale and rent plots) |
| Commercial unit listings | ~5,580 (3,800+ rent, 1,760+ sale) |
| Areas covered | 54 (47 client districts + 7 portal areas); 45 with live data |
| Data sources | Bayut, Dubizzle, Property Finder |
| Monthly running cost | Zero (all services on free tiers) |

## 2. System architecture

How data travels from the property portals to the user's screen.




### Login and security

- Sign in with email and password (Supabase Auth); password reset by email; profile page
- The website reads only the packed snapshot — it never holds database write keys
- The Gemini AI key lives server side only (Vercel function) and is never exposed to browsers
- All secrets live in `.env` files that are never committed to GitHub

# 3. Project structure

The repository is organised into a clear frontend / backend split.

```
├── frontend/              THE WEBSITE (Vite + React) — deployed on Vercel
│   ├── src/               pages, components, styles
│   ├── api/               serverless function for the AI assistant
│   └── public/            logos, icons, landData.json (the data snapshot)
│
├── backend/               THE DATA ENGINE (Python)
│   ├── scrapers/          one file per portal + area matcher + normaliser
│   ├── db/                database schema + Supabase client
│   ├── run_scrape.py      scrape all three portals → Supabase
│   ├── make_snapshot.py   Supabase → frontend/public/landData.json
│   └── requirements.txt   Python dependencies
│
├── docs/                  HANDOVER.md + this guide + prototype archive
├── extras/                optional password reset page
├── setup.bat              one-time PC setup
├── scrape_and_publish.bat ONE CLICK: scrape → rebuild → publish
├── schedule_weekly.bat    turn on the weekly automatic refresh
└── .github/workflows/     free weekly Property Finder cloud refresh
```

| Part | Technology | Where it runs |
|---|---|---|
| frontend | React 19, Vite, Leaflet maps, Web Speech, PWA | Vercel (auto deploy from GitHub main) |
| backend scrapers | Python 3.11+, Patchright (undetected Chrome) | Any Windows PC (one click) + GitHub cloud (PF only) |
| database | Supabase (PostgreSQL + Auth) | Supabase cloud |
| AI assistant | Google Gemini via a Vercel serverless function | Vercel |

# 4. Valuation methodology

Two categories, two honest methods — never mixed.

### Land plots

Bare land is almost never advertised "for rent", so there is no rental market to read directly. The platform therefore uses the standard income approach:

> **Annual land rent = real sale price × yield**
> Yield band 5–8%, default 6.5% (configurable). Sale prices are real listings from the three portals, aggregated per area by median.

### Commercial units

> **Rent is read directly from the market.**
> Offices, shops, warehouses, showrooms, buildings, villas, floors, labour accommodation and factories use the real advertised rents (annualised), plus real sale prices. No yield formula is applied.

### Data quality rules

- **No duplicates:** every listing is stored once, keyed by its URL; re-scrapes update prices in place
- **Medians, not averages,** for per-area statistics — so one mega-priced plot cannot distort an area
- Sizes normalised to square feet; monthly/weekly rents annualised; spam and repeat ads collapsed
- Messy portal locations matched to official areas in English and Arabic, with spelling variants

### Areas with no data — an honest note

Nine districts (Liwara 3, Al Rumailah 3, Mushairef, Al Safia, Mohammed Bin Zayed 1 & 2, Al Raqaib 1 & 2, Al Zahia 2) genuinely have **no listings on any portal** — verified across full fresh scrapes of all three sources. Portals file such listings under parent names (for example "Al Zahya" without a number). The app labels these "no data yet" instead of inventing numbers, and they fill automatically the moment real listings appear.

# 5. The dashboard, page by page

Every page works in English and Arabic and adapts to phones.

| Page | What it shows |
|---|---|
| **Overview** | Rent range for the selected area and type, key figures (lowest / average / highest, 6-month change), top areas chart, AI summary. A Land ↔ Commercial units switch changes the whole view. |
| **Map** | Standard and satellite views; each area is a colour-coded marker (rent level); tap for details. |
| **Listings** | Every listing with size, price and source; sortable; searchable; each row opens the original ad on the portal. |
| **Trends** | Six-month rent trend and distribution for the selected area. |
| **Insights** | Executive stat cards, opportunity map (trend vs price), and a side-by-side comparison of up to three areas. |
| **AI Assistant** | Ask anything about the market by text or voice; answers are grounded in the live data. Also available as a floating chat on every page. |
| **AI Studio** | Rent estimator, ROI calculator and a printable market report. |
| **Profile / Login** | Email + password sign in, password reset by email, show/hide password, account details. |

### Install on a phone (no app store)

- **iPhone:** open the site in Safari → Share → **Add to Home Screen**
- **Android:** open in Chrome → menu → **Install app**

An app icon with the brand logo appears on the home screen and opens full screen like a native app.

# 6. Data operations — refreshing the data

Honest picture: a refresh is a 20–45 minute mostly-unattended run on a normal Windows PC.

### One-time setup on a new PC

1. Install **Python 3.11+** (tick "Add to PATH"), **Google Chrome**, and **Git**
2. Copy the project folder (or `git clone`) and place the `.env` secrets file in the project root
3. Double-click **setup.bat** — installs everything

### Refreshing the data — one click

> **Double-click** `scrape_and_publish.bat`. It scrapes all three portals, rebuilds the data file, and publishes. The live site updates itself about one minute after it finishes.

**What to expect during the run:** Chrome windows open and browse the portals by themselves with polite delays. If Bayut shows a CAPTCHA puzzle, a person clicks it once — the session is then remembered for future runs. This human step cannot be fully removed; it is how the portals protect themselves, and this platform respects it.

### What the run does, step by step

1. **Scrape** — every Ajman land and commercial search on the three portals, page by page
2. **Save** — each listing upserted into Supabase by URL (updates existing, adds new — never duplicates)
3. **Rebuild** — the whole database packed into `frontend/public/landData.json`
4. **Publish** — committed and pushed to GitHub; Vercel deploys automatically

# 7. Automation — schedule and cloud refresh

Two layers: a weekly full refresh on the PC, and a free cloud top-up midweek.

### Weekly automatic refresh (all three portals)

> Double-click `schedule_weekly.bat` once. Every **Sunday 07:00** the PC runs the full refresh by itself. Output is logged to `data\refresh.log`.

- The PC must be **on and logged in** at that time
- Change day/time: edit `/D` and `/ST` inside the file and run it again
- Turn off: `schtasks /Delete /TN "ValuationAI data refresh" /F`

### Free cloud refresh (Property Finder, midweek)

A GitHub Action refreshes Property Finder every **Wednesday** entirely in the cloud — no PC needed (it is the one portal that works without a visible browser). It can also be run on demand: GitHub → **Actions** tab → "Refresh Property Finder data" → **Run workflow**.

**One-time activation:** GitHub repo → Settings → Secrets and variables → Actions → add `SUPABASE_URL` and `SUPABASE_KEY` (values from the `.env` file).

### Why Bayut and Dubizzle cannot run in the cloud

Both use strong anti-bot protection (Cloudflare / DataDome) that blocks invisible browsers and datacenter internet addresses. They require a real Chrome window on a real PC, with an occasional human CAPTCHA click. Any vendor claiming otherwise is overselling — this guide prefers the honest setup that works.

# 8. Accounts, secrets and handover checklist

Four accounts and two secret files define ownership of the platform.

| Account | Holds | Transfer |
|---|---|---|
| GitHub | All code + deploy trigger + cloud refresh | Repo Settings → Transfer ownership |
| Vercel | The live website + AI function + env vars | Project Settings → Transfer; then set `GEMINI_API_KEY` |
| Supabase | Database (all listings) + user logins | Invite new owner → remove old members → **rotate keys** |
| Google AI Studio | Gemini key for the assistant | New owner creates their own free key (aistudio.google.com) |

### Secret files (never in GitHub — hand over separately)

- `.env` (project root) — `SUPABASE_URL` + `SUPABASE_KEY` (the database writer key)
- `frontend/.env` — publishable Supabase key + `GEMINI_API_KEY` (local development only)

### Handover day checklist

1. Client creates GitHub, Vercel, Supabase and Google accounts
2. Transfer the GitHub repo → connect to the client's Vercel → set `GEMINI_API_KEY` → confirm the site deploys (Root Directory = `frontend`)
3. Invite client as Supabase owner → rotate keys → update `.env` on the scrape PC and the GitHub secrets
4. Set up the scrape PC (section 6) and run one refresh together as training
5. Hand over the secret files + this guide; remove old access

# 9. Troubleshooting

The situations that actually happen, and what to do.

| Symptom | Meaning | Fix |
|---|---|---|
| Chrome shows a puzzle / CAPTCHA | Bayut anti-bot check | Solve it in the window once; the run continues and remembers the session |
| A portal logs `0 hits` on every page | The portal changed its website layout | A developer updates that portal's file in `backend/scrapers/` (typically 1–2 hours). Expected a few times per year |
| `Target page … has been closed` at launch | Two scrapes used the same Chrome profile at once | Wait for the other run, or set `SCRAPE_PROFILE_DIR=.pw-profile-2` for the second |
| `git push` rejected | GitHub login on the PC expired | Sign in again when prompted (Windows Credential Manager) |
| Site shows old data after a refresh | Vercel still deploying | Wait 1–2 minutes; hard refresh (Ctrl+Shift+R) |
| AI assistant replies with data cards only | Gemini key missing or minute-limit hit | Check `GEMINI_API_KEY` in Vercel; free tier recovers by itself |
| An area shows "no data yet" | No portal listings matched that district | Genuine market gap (see section 4) — fills automatically when listings appear |

# 10. Maintenance, costs and honest limits

What it takes to keep the platform healthy long term.

## Ongoing maintenance

- **Scraper repairs:** portals redesign their sites a few times a year; each break is a small contained fix in one file. Budget developer time for this, or agree a monthly maintenance retainer — the simplest arrangement for a government client
- **Key hygiene:** rotate Supabase keys at handover and yearly
- **Data cadence:** weekly refresh is a good default; monthly is acceptable for slower decision cycles

## Costs today

| Service | Plan | Cost |
|---|---|---|
| Vercel (website + AI function) | Hobby | Free |
| Supabase (database + logins) | Free tier | Free |
| Gemini (AI assistant) | Free tier | Free (rate limited; paid tier optional later) |
| GitHub (code + cloud refresh) | Free | Free |
| **Total monthly** | | **0 AED** |

## Honest limits, stated plainly

- Refreshing data needs a real PC with a screen for Bayut and Dubizzle; only Property Finder refreshes fully in the cloud
- A human occasionally clicks one CAPTCHA — unavoidable, by portal design
- The platform reports the market as the portals show it; districts absent from the portals show "no data yet" rather than invented figures
- Scrapers are living software: they will need small repairs when portals change

# 11. Glossary

Plain-language definitions of every technical term used in this guide.

| | |
|---|---|
| **Scraping** | Automatically reading listings from a public website with a controlled browser |
| **Portal** | A property listings website — Bayut, Dubizzle, Property Finder |
| **Supabase** | The cloud database that stores every listing and the user logins |
| **Upsert** | "Update or insert" — a re-scraped listing updates its existing row instead of duplicating it |
| **Snapshot (landData.json)** | One packed file holding all per-area statistics and listings; the website reads only this |
| **Vercel** | The hosting service that builds and serves the website automatically on every git push |
| **Yield** | The percentage of a property's value earned as rent per year (land: 5–8%) |
| **Median** | The middle value — used instead of averages so extreme listings cannot distort an area |
| **CAPTCHA** | A "prove you are human" puzzle some portals show; solved once per session by a person |
| **Headless / headed** | A browser without / with a visible window; Bayut and Dubizzle require a visible one |
| **PWA** | "Progressive Web App" — lets the website install to a phone home screen without an app store |
| **Gemini** | Google's AI model powering the assistant; grounded in this platform's own data |
| **GitHub Action** | A free scheduled robot in the cloud; here it refreshes Property Finder weekly |
| **.env file** | A small local file holding secret keys; never uploaded to GitHub |
| **AED/sqft** | Price per square foot in UAE dirhams — the core comparison unit across areas |

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